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ELECTRICAL CONNECTION FOR CHEMICAL MECHANICAL POLISHING CARRIER HEAD

Abstract

A chemical mechanical polishing apparatus has a platen with a top surface to hold a polishing pad, a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, a sensor and/or actuator arranged on the carrier head, and a rotary electrical connection which provides at least two electrical connections between a controller and the sensor and/or actuator. The controller is configured to receive a signal from the sensor and/or actuator and control the carrier head based on the signal.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to chemical mechanical polishing (CMP), and more specifically to a CMP carrier head system having slip ring connections for power and signal.

BACKGROUND

[0002] An integrated circuit is typically formed on a substrate by the sequential deposition of conductive, semiconductive, or insulative layers on a silicon wafer. Planarization of a substrate surface may be required for the removal of a filler layer until a predetermined thickness remains over an underlying layer or an underlying patterned layer is exposed, or to improve planarity for photolithography during fabrication of the integrated circuit.

[0003] Chemical mechanical polishing (CMP) is one accepted method of planarization. This planarization method typically requires that the substrate be mounted on a carrier or polishing head. The exposed surface of the substrate is typically placed against a rotating polishing pad. An abrasive polishing slurry is typically supplied to the surface of the polishing pad. The carrier head provides a controllable load on the substrate to push it against the polishing pad. For example, a membrane can be secured to the carrier head with the outer surface of the membrane contacting the substrate and an inner surface of the membrane forming a pressurizable chamber. A pneumatic line through a drive shaft can provide controllable pressure to the chamber.

SUMMARY

[0004] In one aspect, a chemical mechanical polishing apparatus has a platen with a top surface to hold a polishing pad, a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, a sensor and/or actuator arranged on the carrier head, and a rotary electrical connection which provides at least two electrical connections between a controller and the sensor and/or actuator. The controller is configured to receive a signal from the sensor and/or actuator and control the carrier head based on the signal.

[0005] In another aspect, a chemical mechanical polishing apparatus has a platen with a top surface to hold a polishing pad, a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, a motor, and a drive shaft connecting the motor to the carrier head to rotate the carrier head. The carrier head includes a membrane that forms a plurality of pressurizable chambers, and a sensor and/or actuator configured arranged on or in the carrier head to generate a signal or receive power. The drive shaft has a central passage and a plurality of outer passages surrounding the central passage. At least two wires pass through the central passage to connect to the sensor and/or actuator, and the plurality of outer passages are in pneumatic connection with the plurality of pressurizable chambers. A rotary pneumatic connector provides a plurality of pneumatic connections between the plurality of outer passage and a plurality of individually controllable pressure sources, and a rotary electrical connector provides at least two electrical connections between the at least two wires in the central passage and a controller.

[0006] In another aspect, a chemical mechanical polishing apparatus includes a platen having a top surface to hold a polishing pad, a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, a sensor and/or actuator arranged on or in the carrier head is configured to generate a signal or receive power, a motor, a drive shaft connecting the motor to the carrier head to rotate the carrier head, and a rotary electrical connector. The drive shaft includes a central passage and at least two wires extending through the central passage, and the at least two wires exit the passage through a wall of the drive shaft above the carrier head and are connected to the sensor and/or actuator. The rotary electrical connector provides at least two electrical connections between the at least two wires in the passage and a controller.

[0007] Implementations can include, but are not limited to, one or more of the following potential

advantages.

[0008] Routing electrical cabling through a central passage in the drive shaft can permit the drive shaft to accommodate different cabling needs of different carrier heads. Surrounding the central passage with passages for pneumatic control permits efficient use of the available cross-sectional area. Routing electrical cabling through a side of the draft shaft rather than through an upper housing simplifies construction of the carrier head housing and permits greater flexibility in location of sensors and actuators in the carrier head.

[0009] The details of one or more embodiments set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic cross-sectional side view of an example polishing apparatus.

[0011] FIG. 2 is a schematic cross-sectional side view of a carrier head and drive shaft assembly.

[0012] FIG. 3A is a schematic cross-sectional view through a drive shaft at line 3A-3A in FIG. 2.

[0013] FIG. 3B is a schematic cross-sectional view through a drive shaft at line 3B-3B in FIG. 2.

[0014] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0015] Due to ever increasing requirements of precision in semiconductor manufacturing, there is an ever-increasing need for monitoring and precise control of the polishing process. For control, use of electrically driven actuators, e.g., piezoelectric actors or electric motors, within the carrier head has been proposed. For monitoring, placement of sensors on a carrier head has been proposed. However, existing pneumatic connections cannot carry a signal from a metrology sensor located on the carrier head to an external controller. Although wireless connections have been proposed, this necessitates the use of local power, e.g., batteries, which can run out. Moreover, wireless signals can limit the signal bandwidth available to the controller.

[0016] One technique to establish a wired electrical connection to the carrier head is to route an electrical wire through a slip ring. However, a single electrical connection is not compatible with most sensors and actuators. Even with multiple electrical connections, there is room for improvement when facing actual implementation challenges, e.g., how to route the wires to permit ease of assembly, adaptability for multiple types of carrier heads with different numbers of sensors and/or actuators, etc.

[0017] An approach to address increased needs for direct electrical connections to power sensors and actuators is to use an electrified carrier head system which routes multiple electrical wires to the carrier head through a multi-channel slip ring. The slip ring maintains electrical and signal connections for multiple sensors and/or actuators and allows the carrier head to rotate freely during a polishing process.

[0018] The power and signal channels are maintained by an input-output distribution (IOD) block. The IOD block can be placed before, or after, the slip ring, such as directly on the carrier head. The IOD block can be positioned to facilitate process flexibilities such as multiple signal channels can be combined and transmitted via communication connection such as EtherCAT or DeviceNet (Dnet).

[0019] The wires can be routed through a central passage in a drive shaft up to the slip ring, and the central passage can be surrounded by multiple other passages for pneumatic control of pressurizable chambers in the carrier head. In addition, the wires can exit the drive shaft before the adapter where the housing of the carrier head is mated to the drive shaft. This reduces the need for complex channels and passages to be formed through the housing itself.

[0020] FIG. 1 illustrates an example of a polishing station **120** of a chemical mechanical polishing system **100**. The polishing station **120** includes a rotatable disk-shaped platen **124** on which a polishing pad **130** is situated. The platen **124** is operable to rotate about an axis **125**. For example, a motor **122** can turn a drive shaft **128** to rotate the platen **124**. The polishing pad **130** can be a two-layer polishing pad with an outer polishing layer **134** and a softer backing layer **132**.

[0021] The polishing station **120** can include a supply port, e.g., at the end of a slurry supply arm **139**, to dispense a polishing liquid **138**, such as an abrasive slurry, onto the polishing pad **130**.

[0022] Referring to FIGS. 1 and 2, a carrier head **150** is operable to hold a substrate **110** against the polishing pad **130**. The carrier head **150** can include a flexible membrane **152** having a substrate mounting surface to contact the back side of the substrate **110**, and a plurality of pressurizable chambers **154** to apply different pressures to different zones, e.g., different radial zones, on the substrate **110**. The carrier head **150** can include a retaining ring **160** to hold the substrate below the membrane **152**. In some implementations, the retaining ring **160** may include a lower plastic portion **162** that contacts the polishing pad, and an upper portion **164** of a harder material.

[0023] In operation, the platen is rotated about its central axis **125**, and the carrier head is rotated about its central axis **156** and translated laterally across the top surface of the polishing pad **130**.

[0024] The carrier head **150** is suspended from a support structure **140**, e.g., a carousel or a track, and is connected by a drive shaft **170** to a carrier head rotation motor **142** so that the carrier head can rotate about the axis **156**. Optionally, the carrier head **150** can oscillate laterally, e.g., on sliders on the carousel, by movement along the track, or by rotational oscillation of the carousel itself.

[0025] A representative sensor **180** is shown affixed to an upper surface of the carrier head **150**. Examples of the sensor **180** include an optical sensor, an accelerometer, or a position sensor. The sensor **180** receives power and generates a signal based on the type of sensor.

[0026] An input-output distribution (IOD) block **182** which provides electrical communication between the sensor **180** and the controller **190**. The IOD block **182** provides power, signal communication, or both, to the sensor **180**. The sensor **180** is connected to input-output distribution (IOD) block by one or more wired connections **188**, e.g., a power line **188a** and a ground line, that pass through a rotary electrical connection **184**.

[0027] In some implementations, as shown in FIG. 1, the IOD block **182** is a stationary part (i.e., not rotating with the carrier head **150**). The IOD block **182** connects to the rotary electrical connection **184** which provides electrical connections through the rotary pneumatic connection **144** and the motor **142** to the carrier head **150**, e.g., to the sensor **180**. In particular, the sensor **180** can be connected to input-output distribution (IOD) block by one or more wired connections **188**, e.g., a power line **188a** and a ground line, that pass through the rotary electrical connection **184**.

[0028] In some implementations, as shown in FIG. 2, the IOD block **182** is arranged on the carrier head **150** and the rotary electrical connection **184** provides electrical connection between the controller **190** and IOD block **182** through the drive assembly **172**. In particular, although FIG. 2 illustrates a single line **189** connecting the input-output distribution (IOD) block **182** to the controller **190**, this can be Ethernet-compatible wiring, e.g., an Ethernet cable with five separate wires.

[0029] The rotary electrical connection **184** is a slip ring which maintains individual power and signal connections while revolving around the central axis **156**. Examples of the rotary electrical connection **184** include a drum-type, or a pancake-type slip ring.

[0030] Although the carrier head could have a single sensor or actuator, FIG. 2 illustrates multiple sensors or actuators **180a** in the carrier head **150**. Having an IOD block **182** on the carrier head **150** permits multiplexing of signals to or from the sensors and/or actuators, thus reducing the number of electrical connections needed in the rotary electrical connection **184**. That is, the number of sensors and/or actuators can be larger than the number of electrical connections. This permits a large number of sensors and/or actuators, e.g., ten to a hundred.

[0031] Multiple pneumatic connections extend through the drive shaft **170** to connect the

pressurizable chambers **154** to a rotary pneumatic connection **144**. The rotary pneumatic connection **144** is a multi-channel pneumatic interface having connectors through which individually pressurizable connections extend to individual pressure control valves **146** and a gas pressure source **148**.

[0032] The electrical and pneumatic connections are housed within the drive shaft **170** in separate channels. The drive shaft **170** enclosing the electrical and pneumatic routing channels simplifies the drive assembly **172** and the associated connections between different carrier heads having different numbers of pressurizable chambers thereby permitting uniformity when utilizing different carrier heads. In such examples, the drive assembly **172**, e.g., the drive shaft **170**, is useful with the different carrier heads facilitating interchanging different carrier heads between substrates, or processes.

[0033] FIGS. **3A** and **3B** show cross sections of the drive shaft **170** at two different planes shown in FIG. **2**. The drive shaft **170** includes outer passages **174** which carry pressurized gas from the individual pressure control valves **146** to the individual pressurizable chambers **154** of the carrier head **150**. The outer passages **174** can be evenly spaced around the central axis of the drive shaft **170**. The number of passages **174** can be equal to or greater than the number of pressurizable chambers **154** in the membrane **152**, e.g., some of the passages **174** may not be used by a particular carrier head **150**.

[0034] The drive shaft **170** includes a central channel **178** which carries the electrical connections between the rotary electrical connection **184** and the electrical components on the carrier head **150**, such as the IOD block **182**, or the sensor **180**. The central channel **178** is surrounded by the outer passages **174**.

[0035] FIG. **3B** includes a port **176** through a wall **171** of the drive shaft **170** through which the wired electrical connection from the rotary electrical connection **184** to the IOD block **182** passes. Passing the electrical connection through the port **176** enables a wired connection between the rotary electrical connection **184** and the IOD block **182** which increases available power and signal bandwidth between these components. The carrier head **150** and drive assembly **172** are easier to manufacture having wires on the outer surfaces of the carrier head **150**, e.g., the top, as space within the carrier head **150** is minimal.

[0036] As the drive shaft **170** rotates, the port **176**, IOD block **182**, and wired connection remain in the same rotating reference frame. Such a configuration can be lower cost to assemble than having the wires run through an adapter that connects the drive shaft to the housing of the carrier head. The port **176** is described with reference to the arrangement of FIG. **2** in which the IOD block **182** is on the carrier head **150**, though the same structures, e.g., the passages **174**, channel **178**, and port **176**, are applicable to the arrangement of FIG. **1**, in which the rotary electrical connection **184** is between the sensor **180** and the IOD block **182**. In addition, although only two sensors **180** are illustrated, there could be more sensors and/or actuators.

[0037] In examples in which the sensor **180** is arranged inside the carrier head **150**, it is similarly simpler and more cost effective for manufacturing and servicing to run a through-hole **181** from a point on the outside of the carrier head **150** housing than run the wire **188** through the inside of the carrier head **150**. For example, if an operator desires to remove the carrier head **150** from the drive shaft **170**, it can be simpler to disconnect an electrical connection that is made outside the carrier head, e.g., the connection between cable **189** and the IOD block **182**, before the carrier head **150** is removed from the drive shaft **170** (if wiring runs inside the carrier head into the drive shaft, then it may be necessary to remove and hold the carrier head below the drive shaft to access the wires).

[0038] A carrier head **151** having the sensor **180** installed within a housing **158** is shown in FIG. **2**. The housing **158** includes an opening **186** through which the electrical connection between the IOD block **182** and the sensor **180** transits.

[0039] Embodiments of the invention and all the functional operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware,

or hardware, including the structural means disclosed in this specification and structural equivalents thereof, or in combinations of them. Embodiments of the invention can be implemented as one or more computer program products, i.e., one or more computer programs tangibly embodied in a non-transitory machine-readable storage media, for execution by, or to control the operation of, data processing apparatus, e.g., a programmable processor, a computer, or multiple processors or computers.

[0040] Terms of relative positioning are used to denote positioning of components of the system relative to each other, not necessarily with respect to gravity; it should be understood that the polishing surface and substrate can be held in a vertical orientation or some other orientations.

[0041] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made. Accordingly, other implementations are within the scope of the claims.

Claims

1. A chemical mechanical polishing apparatus comprising: a platen having a top surface to hold a polishing pad; a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process; a sensor and/or actuator arranged on the carrier head; a rotary electrical connection which provides at least two electrical connections between a controller and the sensor and/or actuator, the controller configured to receive a signal from the sensor and/or actuator and control the carrier head based on the signal.
2. The apparatus of claim 1, wherein the rotary electrical connection is a multi-channel rotary electrical connection.
3. The apparatus of claim 1, comprising an input-output distribution (IOD) block, connected to the controller and to the rotary electrical connection.
4. The apparatus of claim 3, wherein the IOD block is connected between the carrier head and the rotary electrical connection.
5. The apparatus of claim 4, wherein the IOD block is mounted on the carrier head.
6. The apparatus of claim 5, wherein the IOD block is mounted on an outside surface of the carrier head.
7. The apparatus of claim 4, comprising a plurality of sensors in the carrier head, and a plurality of wires connecting the plurality of sensors to the IOD block.
8. The apparatus of claim 3, comprising wherein the IOD block is connected between the rotary electrical connection and the controller.
9. The apparatus of claim 1, wherein the carrier head includes a plurality of pressurizable chambers, and the apparatus comprises a drive shaft connecting a motor to the carrier head to rotate the carrier head, the drive shaft including a central passage and a plurality of outer passages surrounding the central passage, wherein at least two wires pass through the central passage to connect to the sensor and/or actuator, and the plurality of outer passages are in pneumatic connection with the plurality of pressurizable chambers.
10. The apparatus of claim 9, wherein the at least two wires exit the passage through a wall of the drive shaft above the carrier head.
11. A chemical mechanical polishing apparatus comprising: a platen having a top surface to hold a polishing pad; a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, the carrier head including a membrane that forms a plurality of pressurizable chambers, the carrier head further including a sensor and/or actuator configured to generate a signal or receive power, the sensor and/or actuator arranged on or in the carrier head; a motor; a drive shaft connecting the motor to the carrier head to rotate the carrier head, the drive shaft including a central passage and a plurality of outer passages surrounding the central passage, wherein at least two wires pass through the central passage to connect to the sensor and/or actuator,

and the plurality of outer passages are in pneumatic connection with the plurality of pressurizable chambers; a rotary pneumatic connector which provides a plurality of pneumatic connections between the plurality of outer passage and a plurality of individually controllable pressure sources; and a rotary electrical connector which provides at least two electrical connections between the at least two wires in the central passage and a controller.

12. The apparatus of claim 11, wherein the rotary electrical connector comprises an electrical slip ring.

13. The apparatus of claim 12, wherein the rotary pneumatic connector is positioned between the rotary electrical connector and the carrier head.

14. The apparatus of claim 13, wherein the rotary pneumatic connector is positioned between the rotary electrical connector and the motor.

15. A chemical mechanical polishing apparatus comprising: a platen having a top surface to hold a polishing pad; a carrier head to hold a substrate against a polishing surface of the polishing pad during a polishing process, the carrier head further including a sensor and/or actuator configured to generate a signal or receive power, the sensor and/or actuator arranged on or in the carrier head; a motor; a drive shaft connecting the motor to the carrier head to rotate the carrier head, the drive shaft including a central passage and at least two wires extending through the central passage, wherein the at least two wires exit the passage through a wall of the drive shaft above the carrier head and are connected to the sensor and/or actuator; a rotary electrical connector which provides at least two electrical connections between the at least two wires in the passage and a controller.

16. The apparatus of claim 15, wherein the carrier head comprises a housing, the sensor and/or actuator is arranged within the housing, and the housing includes an opening through which one of the wires extends through to connect to the sensor.

17. The apparatus of claim 15, wherein the carrier head includes a plurality of pressurizable chambers, and the apparatus comprises a drive shaft connecting a motor to the carrier head to rotate the carrier head, the drive shaft including a central passage and a plurality of outer passages surrounding the central passage, wherein the at least two wires pass through the central passage to connect to the sensor and/or actuator, and the plurality of outer passages are in pneumatic connection with the plurality of pressurizable chambers.

18. The apparatus of claim 17, comprising an aperture extending through the wall of the drive shaft and passing between two of the plurality of outer passages to the central passage, and wherein the at least two wires pass through the aperture.
